Lecture 0

Prerequisites: A Refresher

This course assumes you already know complex numbers, can work with vectors and matrices, and can solve linear systems. Lecture 1 builds on that knowledge rather than re-teaching it. The four topics below are the ones the early lectures lean on hardest.

Complex arithmetic appears on the very first pages of Lecture 1, and from Lecture 7 onward, once we take up inner products, it becomes especially important. Matrix algebra, determinants and Gaussian elimination are needed everywhere: in finding bases of vector spaces, in computing the rank of a linear map, and in analysing the roots of characteristic polynomials.

There is nothing new in this Week 0 lecture — it is here for checking and recalling what you already know. It comes with a companion set of self-check problems (`exercises-0.pdf`) to help you find the gaps. Begin with the problems, then read only those sections of this lecture where you found gaps. Afterwards make sure you can solve all of the self-check problems.

The course also assumes single-variable calculus (including integration by parts), elementary linear ODEs, sequences and series, and that you can expand a function in a Fourier series. The self-check set tests those topics separately; this lecture covers only algebraic readiness and is not a recap of a calculus course.

Learning objectives

By the end of this lecture you will be able to:

- Add, multiply, conjugate, and divide complex numbers, and move freely between Cartesian and polar form.
- Multiply matrices, transpose them, and read off the trace, knowing which operations commute and which do not.
- Evaluate 2×2 and 3×3 determinants and use $\det A \neq 0$ as the test for invertibility.
- Row-reduce a linear system by Gaussian elimination, describe its solution set, and find a basis for the solutions of $Ax = 0$.

1 Complex numbers

Standard quantum mechanics describes microscopic physics by means of complex numbers, so for a physicist a command of complex arithmetic is essential. A *complex number* is written $z = a + bi$ with $a, b \in \mathbb{R}$ and $i^2 = -1$; its *real* and *imaginary* parts are $\Re z = a$ and $\Im z = b$. Addition is componentwise and multiplication follows from $i^2 = -1$ and the distributive law, e.g. $(2+i)(1+3i) = 2 + 6i + i + 3i^2 = -1 + 7i$.

1.1 Conjugate, modulus, division

The *conjugate* of $z = a + bi$ is $\overline{z} = a - bi$, and its *modulus* is

$$|z| = \sqrt{z\overline{z}} = \sqrt{a^2 + b^2}, \qquad \text{so} \qquad z\overline{z} = |z|^2.$$

Conjugation respects the arithmetic, $\overline{z+w} = \overline{z} + \overline{w}$ and $\overline{zw} = \overline{z}\,\overline{w}$, the modulus is multiplicative, $|zw| = |z|\,|w|$, and z is real exactly when $\overline{z} = z$. To divide by a *nonzero* complex number, multiply top and bottom by the conjugate of the denominator, turning it real:

$$\frac{3+2i}{1-i} = \frac{(3+2i)(1+i)}{(1-i)(1+i)} = \frac{3+3i+2i+2i^2}{1^2+1^2} = \frac{1+5i}{2} = \tfrac{1}{2} + \tfrac{5}{2}i.$$

1.2 Polar form and Euler's formula

Every nonzero complex number has a *polar form*

$$z = r\,e^{i\theta} = r(\cos\theta + i\sin\theta), \qquad r = |z|, \quad \theta = \arg z, \tag{0.1}$$

where $r > 0$ and the argument θ is determined modulo 2π. How do we get θ from $z = a + bi$? The equation $\tan\theta = b/a$ fixes θ only up to π, so take $\theta = \arctan(b/a)$ when $a > 0$, add π when $a < 0$, and take $\theta = \pm\pi/2$ (sign of b) when $a = 0$. The most reliable check is to mark the number as a point in the complex plane and read off its quadrant. Zero is written $0 = 0e^{i\theta}$ for every θ, so $\arg 0$ is undefined. The identity $e^{i\theta} = \cos\theta + i\sin\theta$ is *Euler's formula*. For nonzero factors, polar multiplication is almost free—moduli multiply and arguments add modulo 2π,

$$r_1e^{i\theta_1}\cdot r_2e^{i\theta_2} = r_1r_2\,e^{i(\theta_1+\theta_2)}, \qquad \text{so} \qquad \left(re^{i\theta}\right)^n = r^ne^{in\theta}, \qquad n \in \mathbb{Z},$$

(the second identity is *De Moivre's theorem*). Roots reverse this: a nonzero $z = re^{i\theta}$ has exactly n complex n-th roots,

$$z^{1/n} = r^{1/n}\,e^{i(\theta+2\pi k)/n}, \qquad k = 0, 1, \ldots, n-1,$$

obtained by taking the positive real n-th root of the modulus, dividing the argument by n, and then stepping round by $2\pi/n$. (Taking $z = 1$ recovers the roots of unity below.)

Example 0.1 (Powers via polar form). Write $-1 + \sqrt{3}\,i$ in polar form and cube it. Here $r = \sqrt{1+3} = 2$ and the point lies in the second quadrant at $\theta = \frac{2\pi}{3}$, so $-1 + \sqrt{3}\,i = 2\,e^{i2\pi/3}$. Then

$$\left(-1 + \sqrt{3}\,i\right)^3 = 2^3e^{3(i\cdot 2\pi/3)} = 8.$$

Cubing tripled the argument to a full turn, $e^{i\cdot 2\pi} = 1$, and cubed the modulus. In polar form the computation takes a single line, whereas in Cartesian form we would have to expand a binomial. ◁

Remark 0.2. The factor $e^{i\theta}$ is a complex number of modulus 1. In quantum mechanics it is called a pure *phase*. Phases are used to describe quantum interference, which is why standard quantum mechanics works naturally with complex Hilbert spaces; we will meet those in Lecture 13. The solutions of $z^n = 1$, the n-th *roots of unity* $e^{2\pi ik/n}$ with $k = 0, \ldots, n-1$, are phases that occur often in quantum mechanics.

Remark 0.3. Abstractly, complex numbers are defined as ordered pairs (a, b) with $a, b \in \mathbb{R}$, together with the operations $(a, b)+(c, d) = (a+c, b+d)$ and $(a, b)\cdot(c, d) = (ac-bd, ad+bc)$. The inverse of a nonzero complex number (a, b) is $\left(\frac{a}{a^2+b^2}, \frac{-b}{a^2+b^2}\right)$.

2 Matrix algebra

The symbol $\mathbb{F}$ will denote the real or the complex numbers, $\mathbb{R}$ or $\mathbb{C}$. A matrix in $M_{m\times n}(\mathbb{F})$ has m rows and n columns; matrices of the same shape add entrywise and scale entrywise. Below we recall two operations that the course will use constantly.

2.1 Multiplication

The product AB of matrices A and B is defined only when the number of columns of A equals the number of rows of B; if $A \in M_{m\times n}(\mathbb{F})$ and $B \in M_{n\times p}(\mathbb{F})$ then $AB \in M_{m\times p}(\mathbb{F})$ with entries

$$(AB)_{ik} = \sum_{j=1}^{n} A_{ij}B_{jk}.$$

Multiplication is associative and distributes over addition, $(AB)C = A(BC)$ and $A(B+C) = AB + AC$, $(A+B)C = AC + BC$. Multiplying a matrix A by an identity matrix of the right size changes nothing: $AI_n = A = I_m A$ when A is $m \times n$. When A is $n \times n$ this becomes $AI_n = I_n A = A$.

It is important to remember that multiplication can be *non-commutative*:

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix},$$

so in general $AB \neq BA$. An example of non-commutativity in classical physics is the incompatibility of rotations of a ball: two different rotations, performed in different orders, give different results in general. In quantum mechanics observables are represented by matrices, and when two of them fail to commute, the corresponding measurements are incompatible — this is what the Heisenberg uncertainty relation expresses. It is that incompatibility of measurements, rather than non-commutativity itself, which has no classical analogue.

2.2 Transpose and trace

The *transpose* of a matrix, written A^{T}, swaps rows and columns, $(A^{\mathsf{T}})_{ij} = A_{ji}$, and reverses products: $(AB)^{\mathsf{T}} = B^{\mathsf{T}}A^{\mathsf{T}}$. A matrix is *symmetric* if $A^{\mathsf{T}} = A$ and *antisymmetric* if $A^{\mathsf{T}} = -A$; we will meet symmetric and antisymmetric matrices again in Lecture 1.

For a square matrix A, the *trace* is the sum of the entries on the main diagonal,

$$\operatorname{tr} A = \sum_i A_{ii}.$$

The trace is a linear map; we will take up the theory of linear maps in Lecture 3. More generally, if $A \in M_{m\times n}(\mathbb{F})$ and $B \in M_{n\times m}(\mathbb{F})$, then both products AB and BA are square matrices and satisfy the cyclic identity

$$\operatorname{tr}(AB) = \operatorname{tr}(BA), \tag{0.2}$$

even though the matrices AB and BA can differ (and can even have different sizes).

The *inverse* A^{-1}, when it exists, is the square matrix such that $AA^{-1} = A^{-1}A = I$. The inverse also reverses products: for invertible square matrices of compatible size, $(AB)^{-1} = B^{-1}A^{-1}$. In the last two sections we will recall the condition for an inverse to exist and a method for computing it.

Remark 0.4. For complex matrices the especially important operation is the *conjugate transpose* (or Hermitian adjoint) $A^{\dagger} = \overline{A}^{\mathsf{T}}$. It transposes the matrix and conjugates every entry. We will meet the Hermitian adjoint in Lectures 9–10. For now simply note that the complex analogue of the property $A^{\mathsf{T}} = A$ of real symmetric matrices is $A^{\dagger} = A$; such matrices are called *Hermitian.*

3 Determinants

To every *square* matrix A the determinant, written $\det A$ or $|A|$, assigns a scalar that determines whether the matrix is invertible. For 2×2 and 3×3 matrices the formulas are

$$\det\begin{pmatrix} a & b\\ c & d\end{pmatrix} = ad-bc, \qquad \det\begin{pmatrix} a & b & c\\ d & e & f\\ g & h & i\end{pmatrix} = a\begin{vmatrix} e & f\\ h & i\end{vmatrix} - b\begin{vmatrix} d & f\\ g & i\end{vmatrix} + c\begin{vmatrix} d & e\\ g & h\end{vmatrix},$$

The second formula is *cofactor (Laplace) expansion* along the top row. One may expand along any row or column, attaching the sign $(-1)^{i+j}$ to the entry in position (i,j). Computing a determinant becomes simpler if you choose the row or column with the most zeros.

Example 0.5 (A 3×3 determinant). Expanding along the first row,

$$\det\begin{pmatrix} 2 & 1 & 0\\ 1 & 3 & 1\\ 0 & 1 & 2\end{pmatrix} = 2(3\cdot 2-1\cdot 1)-1(1\cdot 2-1\cdot 0)+0 = 2\cdot 5-2 = 8\neq 0,$$

so this matrix is invertible. ◁

3.1 Properties we will often need

Three properties of the determinant come up again and again in this course. The determinant is *multiplicative*: $\det(AB)=\det A\,\det B$. Transposition does not change it: $\det A^{\mathsf T}=\det A$. The determinant of a triangular matrix equals the product of the entries on its main diagonal.

Next we recall how the elementary row operations act. Swapping two rows flips the sign of the determinant; scaling a row by a number c multiplies the determinant by that same c; and adding a multiple of one row to another leaves the determinant unchanged (we rely on this property in the next section). Most importantly,

$$\begin{aligned} A \text{ is invertible} &\iff \det A\neq 0\\ &\iff \text{the columns of } A \text{ are linearly independent}\\ &\iff Ax=0 \text{ only for } x=0. \end{aligned}$$

For a 2×2 matrix the inverse is immediate from the determinant,

$$\begin{pmatrix} a & b\\ c & d\end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b\\ -c & a\end{pmatrix},$$

while in larger sizes row reduction (Section 4) is the practical route.

Remark 0.6 (Looking ahead to Week 5). In this course determinants will be used to find the eigenvalues of linear operators. The *characteristic polynomial* of a square matrix A is $\det(\lambda I-A)$ — a polynomial in λ whose leading coefficient is 1. Its roots, that is, the values of λ for which $\lambda I-A$ is not invertible, are the eigenvalues of the linear operator that A represents. Because you already know determinants, in Week 5 we can define eigenvalues straight away using this very form, unlike Axler's textbook, which approaches them at length.

4 Gaussian elimination

Gaussian elimination is a convenient and widely used method for solving linear systems. From Lecture 2 onward we will apply it to find bases of vector spaces and to compute the rank and the kernel dimension of a linear map. The method consists of three *elementary row operations*: swap two rows, scale a row by a nonzero number, add a multiple of one row to another. These three operations are used to bring a linear system to *reduced row-echelon form* (RREF): all zero rows lie below all nonzero rows; the leading nonzero entry of each nonzero row is a 1, called a *pivot*; each such leading 1 sits strictly to the right of the leading 1 in the row above it; and each leading 1 is the only nonzero entry in its column.

4.1 Solving a system

A linear system is called *consistent* if it has at least one solution, and *inconsistent* if it has none. Write the system as an augmented matrix $[A \mid b]$ and reduce. First check whether the system is consistent: a row $[\,0 \cdots 0 \mid c\,]$ with $c \neq 0$ shows that the system is inconsistent and has no solutions. Otherwise the system is consistent: if there is a pivot in every variable column, the solution is unique; if at least one variable column has no pivot, a *free variable* appears and there are infinitely many solutions.

Example 0.7 (A unique solution). Let us solve the *nonhomogeneous* linear system

$$\begin{aligned} x + 2y + z &= 4, \\ 2x - y + 3z &= 4, \\ -x + y + 2z &= 2, \end{aligned}$$

by Gaussian elimination. First we write down the augmented matrix $[A \mid b]$. Then we reduce it, marking each step with an arrow and writing out the row operations performed. The operations are carried out in the order written, the upper one first; note that on the second and third arrows this order really matters:

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & -1 & 3 & 4 \\ -1 & 1 & 2 & 2 \end{array}\right] \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & -5 & 1 & -4 \\ 0 & 3 & 3 & 6 \end{array}\right] \xrightarrow[R_2 \leftrightarrow R_3]{R_3 \to \frac{1}{3}R_3} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & -5 & 1 & -4 \end{array}\right]$$

$$\xrightarrow[R_3 \to \frac{1}{6}R_3]{R_3 \to R_3 + 5R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right] \xrightarrow[R_1 \to R_1 - R_3]{R_2 \to R_2 - R_3} \left[\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right],$$

We have found that $(x, y, z) = (1, 1, 1)$ is the unique solution of the system. Substituting it into the original equations confirms that all three are satisfied. This is the only example in which we write the elimination steps out one by one; in later examples all the steps are marked with a single $\xrightarrow{\text{RREF}}$ arrow. ◁

4.2 Homogeneous systems and the null space

The *homogeneous* system $Ax = 0$ is always consistent: $x = 0$ is a solution. A homogeneous system has *nontrivial* solutions precisely when a free variable exists. Its solution set, written null A, is the *null space* of A; it is a vector space (you will meet it in Lecture 1). A basis of the null space can be read straight off the RREF — a skill the course will need repeatedly.

Example 0.8 (A basis for the solution space). Take $A = \left(\begin{smallmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{smallmatrix}\right)$. The second row equals the first multiplied by 2, so the RREF is the single equation $x_1 + 2x_2 - x_3 = 0$. The pivot is in the first column, so x_2 and x_3 are the free variables. Put $x_2 = s$ and $x_3 = t$; the equation then gives $x_1 = -2s + t$, and we can write the general solution as a single vector,

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2s + t \\ s \\ t \end{pmatrix} = s \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

In the last step we split the general solution into two terms according to the free variables: the first vector is obtained by taking $s = 1$ and $t = 0$, the second by taking $s = 0$ and $t = 1$. These two vectors are linearly independent (each has a 1 in the position where the other has a 0), so the list $\{(-2, 1, 0),\ (1, 0, 1)\}$ is a basis of the null space, and the space itself is two-dimensional. Here A has *rank* 1 (the number of pivots) and *nullity* 2 (the dimension of the null space), and $1 + 2 = 3$ is the number of columns. This is an application of the rank–nullity theorem of Lecture 3. ◁

These two calculations combine. Suppose the system $Ax = b$ has a solution x_p. Then x is also a solution exactly when $A(x - x_p) = 0$, so the full solution set is

$$x_p + \operatorname{null} A = \{x_p + v : Av = 0\}.$$

To find the solution set, first row-reduce $[A \mid b]$, set every free variable to 0 and read off the particular solution x_p. Then take $b = 0$ and read off a basis of the null space in the same way as above. The answer is x_p plus an arbitrary linear combination of those basis vectors. This is exactly the form of answer the tutorial problems will ask for.

Example 0.9 (The solution set of a nonhomogeneous system). Take the matrix of the previous example, $A = \left(\begin{smallmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{smallmatrix}\right)$. Let $b = \left(\begin{smallmatrix} 1 \\ 2 \end{smallmatrix}\right)$. Reducing the augmented matrix $[A \mid b]$ we obtain a single equation:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 4 & -2 & 2 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \implies x_1 + 2x_2 - x_3 = 1.$$

This system is consistent, and x_2 and x_3 are the free variables. Assigning zeros to the free variables, $x_2 = x_3 = 0$, gives the particular solution $x_p = (1, 0, 0)$. We already found a basis of the null space in the previous example, so the full solution set is

$$x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix},$$

where s and t are arbitrary numbers. ◁

Remark 0.10 (Inverses by row reduction). Row reduction also lets us find an inverse matrix. First write down the augmented block $[A \mid I]$ and reduce it. If A is invertible, the left half becomes I and the right half becomes the inverse we are looking for, A^{-1}. If a zero row appears on the left, A is not invertible. This method complements the determinant test $\det A \neq 0$ of Section 3.

Summary of main concepts

The four skills above will be especially important in this course. From complex arithmetic you will need the conjugate, the modulus, division by a nonzero number, and the polar form $re^{i\theta}$ of nonzero numbers. From matrix algebra you will need multiplication (non-commutative!), transpose, the Hermitian adjoint, and trace. Determinants give the simple invertibility test $\det A \neq 0$ and, in Week 5, the characteristic polynomial $\det(\lambda I - A)$. We will solve linear systems by Gaussian elimination, and a basis read off the RREF of $Ax = 0$ will give the null spaces, ranks and dimensions that Lectures 1–4 turn into abstract structure. If the self-check problems expose a gap, repair it before Lecture 1.

- **Complex arithmetic**—$\bar{z}$, $|z| = \sqrt{z\bar{z}}$, division by a nonzero number using its conjugate, and nonzero $z = re^{i\theta}$ with arguments defined modulo 2π.

- **Matrix algebra**—$(AB)_{ik} = \sum_j A_{ij}B_{jk}$ (and $AB \neq BA$); $AI_n = I_m A = A$ for $A \in M_{m\times n}$; transpose A^{T}; Hermitian adjoint $A^\dagger$; trace $\mathrm{tr}(AB) = \mathrm{tr}(BA)$ for compatible rectangular factors.
- **Determinants**—2×2 and 3×3 by cofactors; $\det(AB) = \det A \det B$; $\det A \neq 0 \iff$ invertible $\iff$ columns independent.
- **Gaussian elimination**—complete RREF; check consistency first, then pivots and free variables; the null space of $Ax = 0$ and a basis for it; rank as the pivot count.

The companion set of self-check problems is in the file `exercises-0.pdf`. Begin with it; where you find gaps, review the corresponding sections of this lecture and make sure you can solve all of the problems.